XIAMEN UNIVERSITY MALAYSIA

ONLINE FINAL EXAMINATION

Course Code:	EEE313			
Course Name:	Semiconductor Physics and Device Processes			
Question Paper Setter:	MohammadMahdi Ariannejad			
Academic Session:	2021/09	Question Paper:	A ☑	B ☐
Total No. of Pages:	3	Time Allocated:	2 hours	
Additional Materials:	Calculator, Appendix			
Apparatus Allowed:	-			

INSTRUCTIONS TO CANDIDATES

1. This paper consists of 3 Questions, you need to answer all questions.
2. Please follow the requirement of each section and type in all the corresponding answers on the downloadable answer book provided.
3. Communication between candidates in any means is forbidden. Answers must be entirely individual candidate's independent effort. If you are found sharing your solutions with other candidates, or suspected of doing so, you would be penalized accordingly.

(Student ID: _____ Full Name: _____)

Question 1: (40 marks)

A) Consider an ideal silicon pn junction diode with the geometry shown in Figure 1. The doping concentrations are $N_a = 8 \times 10^{17}$ cm^{-3} and $N_d = 1.7 \times 10^{17}$ cm^{-3}, and the minority carrier lifetimes are $\tau_{n0} = 4 \times 10^{-8}$ s and $\tau_{p0} = 9 \times 10^{-8}$ s. The cross-sectional area is $A = 8 \times 10^{-5}$ cm^2. Estimate:

(i) The ideal reverse-saturation current due to holes, **(CLO1-PLO1, C2) [5 Mark]**

(ii) The ideal reverse-saturation current due to electrons, **(CLO1-PLO1, C2) [5 Mark]**

(iii) The hole concentration at $x = x_n$ for $V_a = 0.8V_{bi}$, **(CLO1-PLO1, C2) [6 Mark]**

(iv) The electron current at $x = x_n$ for $V_a = 0.8\ V_{bi}$, **(CLO1-PLO1, C2) [4 Mark]**

(v) The electron current at $x = x_n + (1/2)\ L_p$ for $V_a = 0.9\ V_{bi}$. **(CLO1-PLO1, C2) [6 Mark]**

(D_p =12 cm^2/s, D_n=20 cm^2/s)

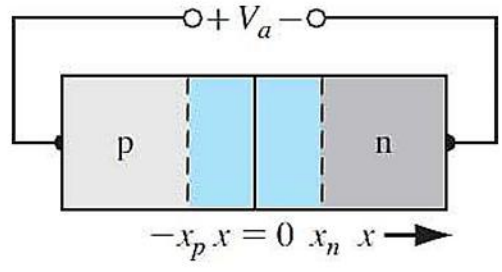


Figure 1.

B) Consider a W–n-type silicon Schottky barrier at $T = 300$ K with $N_d = 8 \times 10^{16}$ cm^{-3}. The barrier height is 0.73 V.

(i) Estimate V_{bi}, x_n, and $|E_{max}|$ for $V_R = 12$ V and $V_R = 8$ V.

(CLO1-PLO1, C2) [8 Marks]

(ii) Using the values of $|E_{max}|$ from part (i), estimate the Schottky barrier lowering parameters $\Delta\phi$ and x_m. **(CLO1-PLO1, C2) [6 Marks]**

($N_c = 2.7 \times 10^{19}$ cm^{-3}) ($\epsilon_s = (11.7)\left(8.85 \times 10^{-14}\right)$)

Question 2: (30 marks)

Chung is interested to learn about Silicon based transistors. In his research he learnt about the pnp bipolar transistor with consideration of room temperature. He also found some information related to the cross sectional of B-E that has the $A_{BE} = 8 \times 10^{-4}$ cm^2 , neutral base width of $x_B = 0.80$ μm, a neutral emitter width of $x_E = 0.60$ μm, and uniform doping concentrations of $N_E = 7 \times 10^{17}$ cm^{-3}, $N_B = 10^{18}$ cm^{-3}, and $N_C = 10^{15}$ cm^{-3}. Other transistor parameters are $D_B = 18$ cm^2/s, $D_E = 25$ cm^2/s, $\tau_{E0} = \tau_{B0} = 8 \times 10^{-8}$ s, and $\tau_{C0} = 2 \times 10^{-6}$s. The transistor is biased in the forward-active mode and the recombination factor is $\delta = 0.885$. Calculate the collector current for :

(i) $V_{EB} = 0.880$ V **(CLO2-PLO2, C4) [14 Marks]**

(ii) $I_B = 1.80$ μA **(CLO2-PLO2, C4) [12 Marks]**

(iii) $I_E = 725$ μA. **(CLO2-PLO2, C4) [4 Marks]**

Question 3: (30 marks)

An ideal MOS capacitor with an n^+ polysilicon gate has a silicon dioxide thickness of t_{ox}= 18 nm on a p-type silicon substrate doped at $N_a = 10^{18}$ cm^{-3}. Design the capacitance C_{ox}, C'_{FB} , C'_{min} , and C' (inv) at

(i) f = 1 Hz, **(CLO3-PLO3, C6) [8 Marks]**

(ii) f = 1 MHz. **(CLO3-PLO3, C6) [6 Marks]**

(iii) Derive V_{FB} and V_T values. **(CLO3-PLO3, C6) [6 Marks]**

(iv) Elaborate the 12 stages for fabrication of the MOSFET with the CMOS technology (you need to draw 12 steps with short explanation for each step).

(CLO3-PLO3, C6) [10 Marks]

($\varepsilon_{ox}= (3.9)(8.85\times10^{-14})$, ($\varepsilon_s =(11.7)(8.85\times10^{-14})$), $\varphi_{ms} \cong -5.10\,V$)

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